

Coalition for Advancing Research Assessment (CoARA) action plan from the Association for Interdisciplinary Meta-Research and Open Science (AIMOS)

November 2024

Introduction

AIMOS is a scholarly society whose mission is to improve the quality of scientific research. We have members from across the globe and cover many fields of research, including archaeology, psychology, ecology, economics, history, and medicine. We work in the field of meta-research, also known as “research on research”, which includes studying research assessment systems.

Many of our members are deeply concerned about poorly designed research assessments which are negatively influencing science, including the obsession with journal impact factors and/or ranking, and a focus on quantity at the expense of quality.

The AIMOS board approved the signing of the Agreement on Reforming Research Assessment in March 2023.

How we have assessed researchers

We have awarded travel scholarships (AUD \$1000) and top-ups scholarships (AUD \$2500) to PhD and Masters students working in meta-research.

Top-up scholarship applicants were asked to provide a short qualitative statement covering their research project and their progress to date. We deliberately kept the application short (just 200 words) as lengthy applications create a barrier to students with commitments outside their research, including caring responsibilities and/or paid work. We believe that a short application will help increase diversity and reduce application costs.

Peer reviewers were asked to assess the top-up applicants as eligible or not. All applicants judged as eligible were then entered into a draw and chosen by lottery. Using a lottery (after initial peer review) is an ideal method for increasing diversity as it removes human biases. Some applicants told us that they would not have applied without the lottery, providing anecdotal evidence that it encourages submissions. However, we cannot rule out that others did not apply because of the lottery, although we have not received any complaints about the process.

Our lottery is stratified so winners were not all the same gender or from the same country. This guarantees diversity in our winners.

Being rejected by a lottery is less painful for early career researchers, as being rejected by peer reviewers can feel personal. However, in highly competitive systems where applicants are ranked using peer review, the winners and losers are somewhat random. Using a lottery also helps reduce perceived conflicts of interest in our relatively small

community. For example, if a PhD student being supervised by the current president won a top-up scholarship using traditional peer review, then that could easily be perceived as favouritism.

Some quotes from top-up scholarship recipients include:

"The top-up scholarship...has been extremely helpful in paying for basic things like groceries, bills and rent while we are experiencing a cost-of-living crisis. It meant that I didn't have to do as much casual teaching to supplement my stipend income and could spend those hours working on my phd instead."

"The AIMOS Top-up Scholarship was a great opportunity for me to talk about my thesis project and to interact with other people on my research topic. It was also a chance to share and discuss some of my results to the meta-research community. Finally, the research grant allowed me to participate in seminars and courses that have been very useful for my project."

"I first want to express my gratitude for being awarded the AIMOS Top-up scholarship. The top-up scholarship provided the much-needed time (outside of my PhD program) to assess the current state of Big Teams Science. Findings that provided insight into the steps Big Team Science organizations and projects still need to take to continue to strive for inclusive and diverse research and researchers."

We give our top-up winners a prominent position in our annual conference, as they are each given a 15-minute presentation without the need to submit an abstract for peer review, compared with the usual 5 minutes for accepted oral presentations.

Government policy

We aim to influence policies on research assessment. We provided submissions to the:

- Australian Government Accord discussion (to "drive lasting and transformative reform in Australia's higher education system") – March 2023.
- Australian Government Policy Review of the National Competitive Grants Program – May 2024.

In these submissions we discussed the problems of using simplistic metrics to assess researchers. We advocated for the principles of research that is robust and transparent.

Future plans

Our instructions to scholarship applicants did not provide any specific text about the use of quantitative indicators. We did not see any use of metrics in the applications we received, however for future rounds we could specifically state that we are a signatory of COARA and provide guidance around not using journal impact factors or the h-index.

We will continue to provide submissions to government inquiries that concern research assessment and/or research integrity.